PROBABILITY AND STATISTICS

Dr. Öğr. Üyesi Sadra Mousavi

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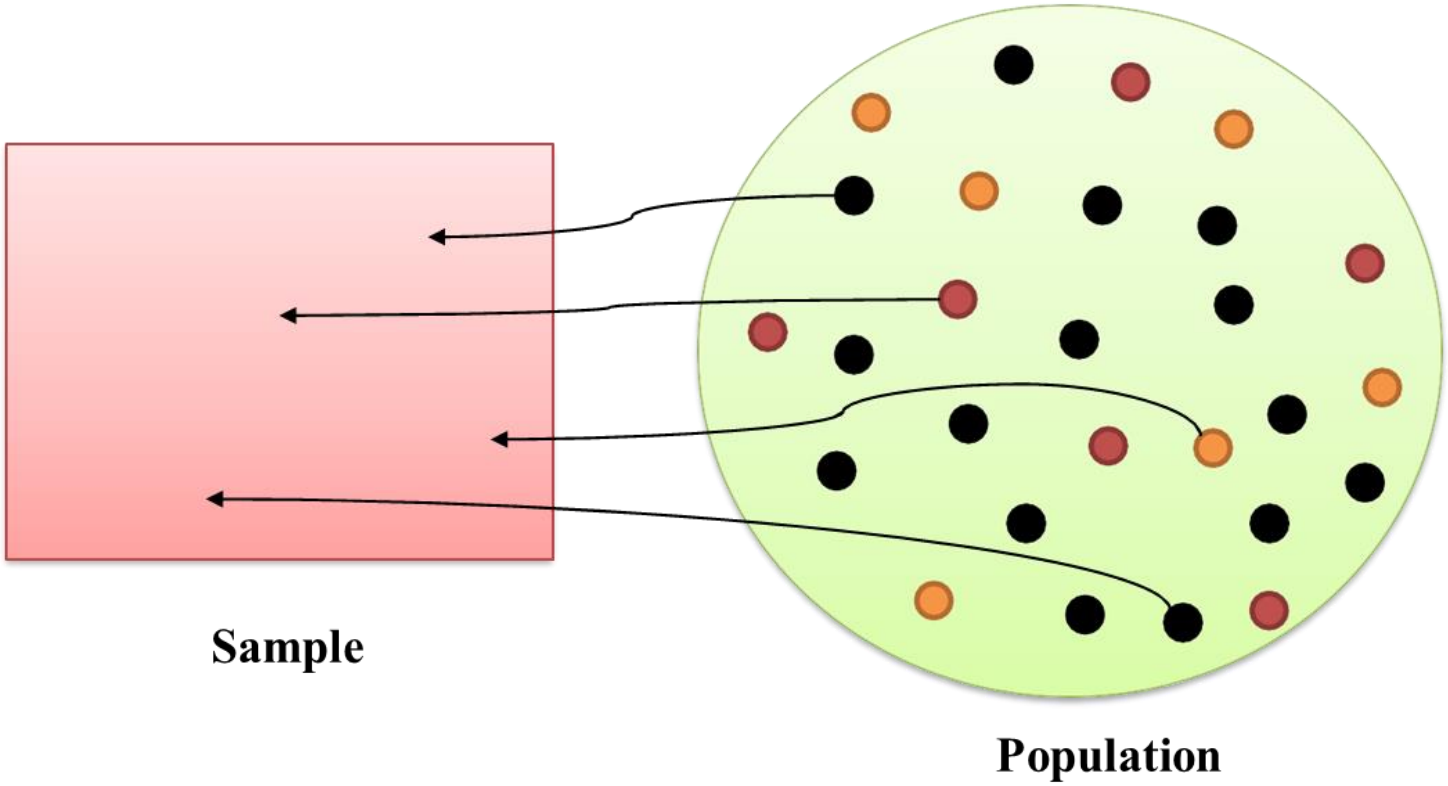
Statistics

- **Definitions:**
- **A population** is a collection of all elements that possess a characteristic of interest.
- Populations can be *finite* or *infinite*.
- A population where all the elements are easily countable may be considered as finite, and a population where all the elements are not easily countable as infinite.
- A **portion** of a population selected for study is called a sample.

Statistics



Statistics

- **Statistics:**
- Statistics concerns data; their collection, analysis, and interpretation.

- Descriptive statistics concerns the summarization of data. We have a dataset and we would like to describe the data set in multiple ways.

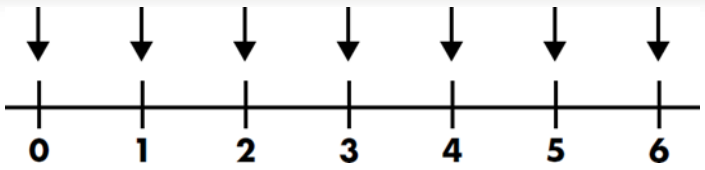
- Descriptive statistics consists of the collection, organization, summarization, and presentation of data.

Statistics

Types of Data

- **Quantitative**
  - *Represent numerical quantities*
    - ***Discrete**:* take values in a finite or countably infinite set of numbers.
      Examples: counts, number of arrivals, or number of cars.
    - ***Continuous***: take values in an interval of numbers.
      Examples: height, weight, length, time.
- **Qualitative**
  - *Do not represent numerical quantities (e.g., gender, color, …)*
- **Logical**
  - *True/False*
- **Missing**
  - *Data that should be there but are not*
- **Other Types**

Statistics



- **Discrete Quantitative Data:**
- Example:
- 12 people were asked about their own cars and the results were recorded as follows:

$$\mathbf{2, 0, 4, 2, 2, 3, 2, 2, 4, 2, 2, 2}$$

Car(s)

- The measuring: *numerical*.
- **Continuous Quantitative Data:**
- Example:
- 10 people were asked about their height and t

$$\mathbf{175, 184, 160, 170, 173, 170, 175, 165, 171, 176}$$

cm

- The measuring: *numerical*.
- **Continuous quantitative data is a scale of measurement which can consist of numbers other than whole numbers like fractions.**

Statistics

Representation Using

- **Tables**
- **Graphs**
- **Numerical Measure**
  - *Only on the* ***Quantitative*** *Data*

Statistics

- Example1:
- 11 people were asked about their gender and the results were recorded as follows:

$$\boldsymbol{M}, \boldsymbol{F}, \boldsymbol{M}, \boldsymbol{M}, \boldsymbol{M}, \boldsymbol{F}, \boldsymbol{F}, \boldsymbol{M}, \boldsymbol{M}, \boldsymbol{M}, \boldsymbol{M}$$

- What is the type of this data ?
- Construct the frequency table?
- Show them with bar graph and pie chart

Statistics

- Example:
- 12 people were asked about their own cars and the results were recorded as follows:

$$\mathbf{2, 0, 4, 2, 2, 3, 2, 2, 4, 2, 2, 2}$$

Car(s)

- What is the type of this data?
- Construct the frequency table?
- Draw the bar graph and pie chart

Statistics

- Example:
- 10 people were asked about their height and the results were recorded as follows:

175, **184**, **160**, **170**, **173**, **170**, **175**, **165**, **171**, **176**

Cm

- What is the type of this data?
- Construct the frequency table?
- Draw the histogram of height for this data?
- Find the mean, median and the mode?

Statistics

- **Histogram**
- A histogram is a graphical tool consisting of bars placed side by side on a set of intervals (classes, bins, or cells) of equal width. The bars represent the frequency or relative frequency of classes. The height of each bar is proportional to the frequency or relative frequency of the corresponding class.